

SMART INDIA HACKATHON 2026



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HACKATHON
2026

- **Problem Statement ID – PS038**

- **Problem Statement Title:**

Explainable AI for Diabetic Retinopathy Screening in Rural India

- **Theme-** MedTech / BioTech / HealthTech

- **PS Category-** Software

- **Team ID-**

- **Team Name:** CHANCE TO CHANGE





THE PROBLEM:

India has ~80 million diabetic patients (2nd highest globally) with ~20% susceptible to DR (leading cause of preventable blindness). Early Screening can prevent 90% vision loss but India has 1 ophthalmologist per 100,000 rural population. Mass manual screening becomes infeasible.



WHY TO IMPLEMENT?

Provide access to early screening can prevent 90% vision loss despite ~1 ophthalmologist per 100,000 rural population ratio. Protect ~20% susceptible population.



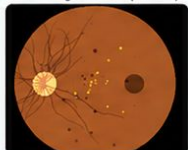
WHAT TO DO?

Use Classical AI approaches to perform mass-diagnostics over large group of people.

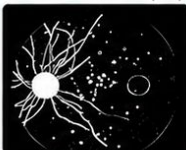


PROPOSED SOLUTION SCHEMATIC

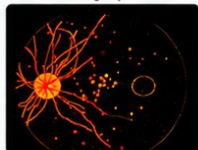
1. Working resolution (872x872)



2. Vessels -- matched filter (7.1%)



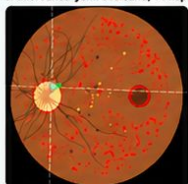
3. Calibre: beading/majorbranch/minor



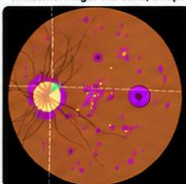
4. Fine-scale excess nv_score 0.74



5. Microaneurysm: 515 cand, 14 kept



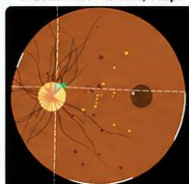
6. Haemorrhage: 148 cand, 0 kept



7. Hard exudate: 38 cand, 2 kept



8. Cotton-wool: 40 cand, 4 kept



REQUIRED SPECIFICATION OF IMPLEMENTED SOLUTIONS:



Image Quality Assessment and Enhancement



Retinal Structure Segmentation: clear mapping (localisation, segmentation, classification & detection)



Severity Grading: scale 0-4; sensitivity (>90%), specificity (>85%); referable DR (Level 2+)



Explainability: Grad-CAM Attention & Automated Reports

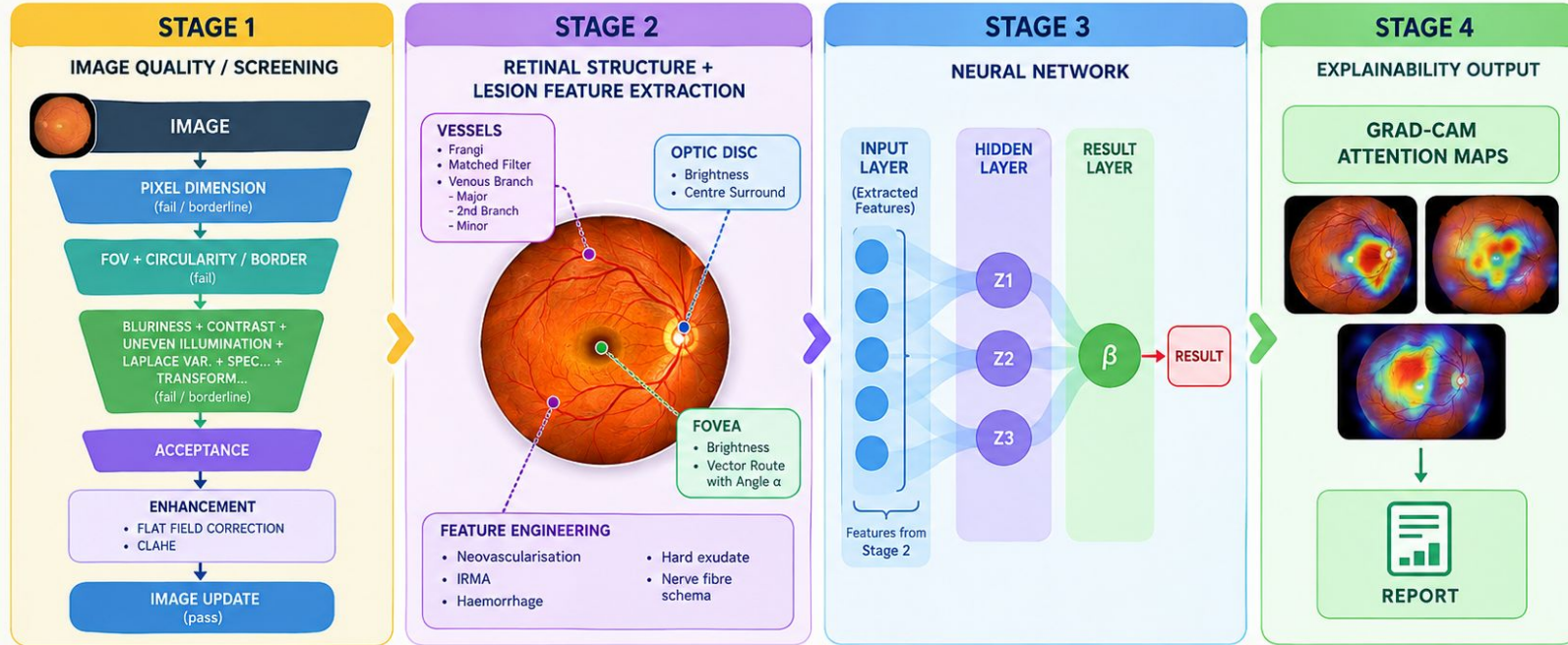


Simulink Workflow



PROBLEMS ADDRESSED:

- Cost
- Speed
- Edge Cases
- Uncertainty Handling
- Clinical Rule Override
- Appropriate Algorithms
- Multi-level Cascade
- Local over Cloud
- Patient Privacy
- Explainable Reports



FOR IN-DEPTH



[CLICK HERE](#)

TECH STACK



MATLAB



Vercel



Vite



React



Node.js



MongoDB



TECHNICAL FEASIBILITY

Can we build it?

01

Field Tested Software



- 1000+ patients reports in ~1 hour on a standard laptop
- Laptop native
- No expensive hardware
- Lightweight design
- Robust edge-case handling

02

Production Oriented: Implementable & Scalable



- Designed to be scalable across thousands of procedures
- Large populations
- High-throughput pipeline
- Easy deployment & integration

03

Hybrid CV + RNN Approach



- Reduce repeated image processing
- Running-time optimisation
- Metrics from RNN
- Explainability
- Waterfall + fallback methods (multi-level cascade)



MARKET FEASIBILITY

Can it scale & be adopted?

01

Large Screening Need



- Large screening need
- Fast + low cost
- Growing diabetic population
- High-volume routine screening
- Cost-effective scalable solution

02

Trust-driven Adoption



- Explainable maps
- Clinician verification
- Human-in-the-loop review
- Clinician trust

03

Explainable AI + Grad-CAM



- Grad-CAM attention maps
- Easily surveyable reports
- Understandable by patients
- Quick verification by doctors



RISK

What can go wrong?



False Negatives



Regular Classical ML Approaches



Missed Clinical Rule Override



Cloud Cost & Privacy



Regulatory / Clinical Safety



SOLUTION

How do we handle it?



High-sensitivity thresholds + cascade verification + clinical overrides



Speed, cost & edge-case optimisation



Definite rules over probabilistic networks



Local processing on simple hardware



Clinical validation + clinician oversight



RESULT

What do we achieve?



Near-zero false negatives
More accurate detection



Superior combined workflow
Faster, efficient, reliable



Correct results even when algorithms miss
Rule-based safety net



No cloud + patient data safe
Privacy and cost control



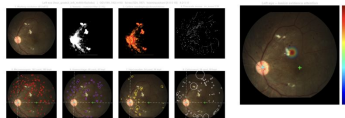
Protocols + standardisation
Safe, consistent, scalable

REPORT

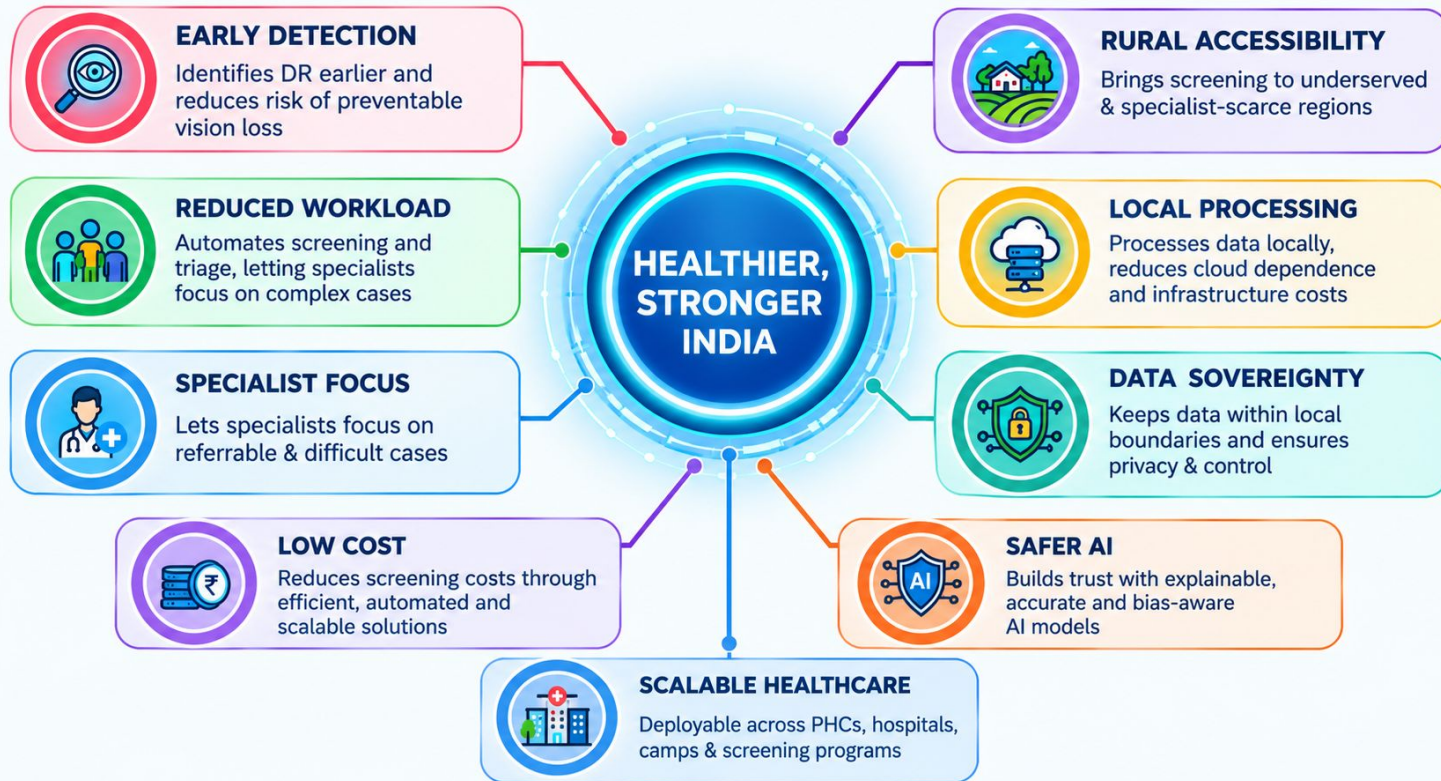
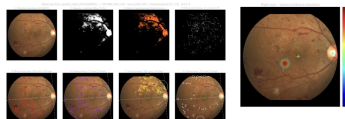
Retinal Screening Report
FLAGGED FOR REVIEW — worst eye LEFT: Proliferative DR (level 4),
referable probability 95%

Patient
 Patient ID RS-3ABA3FC7
 Age 43
 Sex Male
 Diabetes Type 1
 Years since diagnosis 23
 HbA1c 6%
 Blood pressure 130/85

Left eye — Proliferative DR (level 4)
 Image four_grade3_left_4c6b10a3a6a
 Referable (level 2+) YES
 Calibrated probability 0.95
 Grade score 2.827
 Image quality (Stage 1) pass
 Microaneurysms 160
 Hemorrhages 45 (per quadrant 6/0/8/31)
 Hard exudates 54
 Cotton-wool spots 99
 Neovascularisation score 2.29
 DME flag (exudate near fovea) YES
 Criterion Microaneurysms / hemorrhages present



Right eye — Proliferative DR (level 4)
 Image four_grade3_right_237c078a006
 Referable (level 2+) YES
 Calibrated probability 0.99
 Grade score 3.900
 Image quality (Stage 1) pass
 Microaneurysms 147
 Hemorrhages 71 (per quadrant 28/0/14/2)
 Hard exudates 103
 Cotton-wool spots 37
 Neovascularisation score 2.41
 DME flag (exudate near fovea) YES
 Criterion Microaneurysms / hemorrhages present



◆ Dataset

APTOS 2019 Blindness Detection:

<https://www.kaggle.com/c/aptos2019-blindness-detection>

IDRiD (Indian Diabetic Retinopathy Image Dataset):

<https://ieeedataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid>

DRIVE Messidor-2 (Vessel Extraction):

<https://drive.grand-challenge.org/Messidor-2/>
<https://www.adcis.net/en/third-party/messidor2/>

◆ Books & Papers

Gulshan et al. (2019) — *Performance of a Deep-Learning Algorithm vs Manual Grading for Detecting Diabetic Retinopathy*, JAMA Ophthalmology.

Krause et al. (2018) — *Grader Variability and the Importance of Reference Standards for Evaluating Machine Learning Models for Diabetic Retinopathy*, Ophthalmology.

Selvaraju et al. (2017) — *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*, ICCV.

Guo et al. (2017) — *On Calibration of Modern Neural Networks*, ICML.

Jazbec et al. (2024) — *Fast yet Safe: Early-Exiting with Risk Control*, NeurIPS 2024.

Sudre et al. (2024) — *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging (UNSURE)*, MICCAI Workshop.